Soil Digital Twin System For Smart Farming Applications Using IOT & Machine Learning

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Abstract—The Soil Digital Twin System for Smart Farming Applications using IoT & Machine Learning is an intelligent agricultural framework designed to improve soil monitoring and farm management through data-driven decision-making. The system utilizes strategically placed IoT sensors to capture critical soil parameters such as moisture and temperature while avoiding dense sensor deployment, thereby reducing hardware costs, maintenance, and energy consumption. The collected sensor data is pre-processed to remove noise and inconsistencies, ensuring accurate inputs for analysis. Using this refined data, a virtual representation of the physical soil environment, known as the Soil Digital Twin, is created to continuously reflect real-time soil conditions. Machine learning techniques are integrated to analyse both historical and live data, enabling the system to understand soil behaviour patterns and dynamically update soil parameters. Based on these insights, the system provides reliable information to support effective irrigation planning, crop selection, and soil management. By combining optimized sensing, Digital Twin technology, and intelligent analytics, the proposed system reduces manual intervention, enhances resource utilization, and promotes scalable, cost-effective, and sustainable smart farming practices.

Keywords: Soil Digital Twin, Smart Agriculture, IoT, Soil Monitoring, Machine Learning, Automated Irrigation

I.INTRODUCTION

The rapid growth of the global population and the increasing demand for food have placed immense pressure on the agricultural sector to adopt smarter and more efficient farming practices. Soil health plays a crucial role in determining crop productivity, yet traditional soil monitoring methods rely on manual sampling and periodic laboratory testing, which are time-consuming, labour-intensive, and incapable of providing real-time insights. Farmers often struggle to make informed decisions regarding irrigation, fertilization, and crop selection due to the lack of continuous soil data, resulting in reduced yields and inefficient resource utilization. With the advancement of modern technologies, the integration of Internet of Things (IoT), Machine Learning (ML), and Digital Twin systems has emerged as a promising solution to transform conventional agriculture into data-driven precision farming [1]–[3].

Digital Twin technology enables the creation of a virtual replica of physical agricultural environments, allowing dynamic monitoring and predictive analysis of soil conditions. By combining IoT sensors with intelligent analytics, it becomes possible to continuously track critical soil parameters such as moisture, temperature, humidity, and nutrient levels [2], [4]. Machine learning algorithms further enhance the system by analysing historical and real-time data to generate predictive insights, helping farmers optimize irrigation schedules, prevent soil degradation, and improve crop health [3]. Unlike traditional monitoring approaches that provide only static observations, a Soil Digital Twin provides a dynamic and continuously evolving model capable of simulating soil behaviour under varying environmental conditions [5].

The proposed **Soil Digital Twin System for Smart Farming Applications using IoT & Machine Learning** focuses on developing an intelligent framework that bridges the gap between physical farmland and its digital counterpart. The system captures real-time sensor data and synchronizes it with a virtual soil model to support accurate monitoring and proactive decision-making.

Advanced analytics enable early detection of unfavorable soil conditions, allowing timely interventions that enhance productivity while minimizing water and fertilizer wastage. Additionally, the architecture is designed to reduce excessive sensor dependency by leveraging predictive ML models, thereby improving scalability and cost-effectiveness for large-scale agricultural deployment.

Furthermore, the system incorporates automated data visualization and alert mechanisms to assist farmers in understanding soil trends without requiring extensive technical expertise. By presenting actionable recommendations, the platform supports sustainable farming practices and promotes efficient resource management. The integration of AI-driven predictions with Digital Twin technology not only improves operational efficiency but also strengthens resilience against climate variability. As smart agriculture continues to evolve, this approach represents a significant step toward building adaptive and intelligent farming ecosystems capable of addressing future food security challenges while ensuring environmental sustainability.

II. RELATED WORK

Several research efforts have explored the application of Digital Twin technology, IoT, and artificial intelligence to enhance agricultural monitoring and farm management. Verdouw et al. (2021), in *Computers and Electronics in Agriculture*, proposed a Digital Twin framework for smart farming that integrates IoT sensors with a virtual farm model to improve monitoring and operational visibility. The study demonstrated that digital representations can significantly enhance farm management; however, the approach required dense sensor deployment, leading to increased infrastructure cost.

Similarly, Liu et al. (2022), in *IEEE Access*, introduced a Digital Twin framework specifically for soil monitoring using sensor data combined with simulation techniques to track soil conditions accurately. Although the system achieved reliable monitoring, it faced challenges related to scalability and lacked advanced AI-based learning capabilities for predictive analysis.

Patil et al. (2020), published by Springer, developed a smart agriculture system that utilized real-time sensor data along with machine learning models to improve soil and crop monitoring. While the system enhanced decision-making through intelligent analytics, it did not incorporate Digital Twin technology and depended heavily on multiple sensors.

Kumar et al. (2019), in an IEEE publication, designed an IoT-based soil moisture monitoring system using distributed sensors to provide real-time detection of soil conditions. Despite its effectiveness in monitoring moisture levels, the system did not support historical data modelling and involved high maintenance due to extensive sensor usage.

Furthermore, Grieves et al. (2022), in *MDPI Sensors*, presented a comprehensive review of Digital Twin technology in agriculture, highlighting its potential to transform farm management through virtual modelling and real-time insights. However, the work remained largely conceptual and lacked practical implementation strategies. From the reviewed studies, it is evident that existing solutions either rely on sensor-intensive architectures, lack intelligent learning mechanisms, or remain theoretical without scalable deployment models.

These limitations emphasize the need for an integrated system that combines optimized sensor placement, AI-driven soil behaviour analysis, and a dynamic Digital Twin model. The proposed **Soil Digital Twin System for Smart Farming Applications using IoT & Machine Learning** addresses these gaps by minimizing sensor dependency while enabling accurate, scalable, and data-driven soil monitoring for precision agriculture.

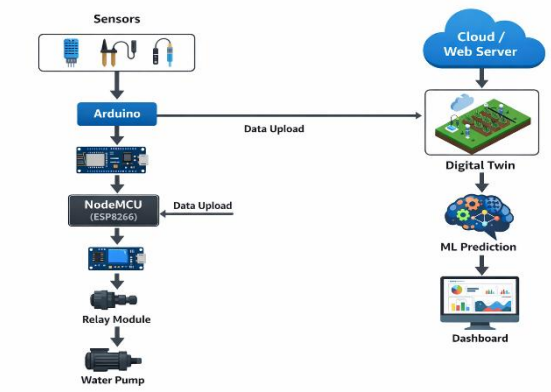
III. PROPOSED SYSTEM WORKFLOW

The proposed **Soil Digital Twin System for Smart Farming Applications using IoT and Machine Learning** is designed to provide intelligent soil monitoring, predictive analytics, and automated irrigation support for modern agriculture. The system integrates multiple soil sensors, microcontrollers, communication modules, and a web-based digital twin platform to create a real-time virtual representation of agricultural land.

The overall architecture consists of three major modules:

- Soil Data Acquisition Module
- Processing and Communication Module
- Digital Twin Visualization and Decision Support Module

Initially, soil parameters are captured using distributed sensors deployed across the agricultural field. The sensed data is transmitted to the microcontroller for processing and forwarded to a web server where the Digital Twin model is maintained. The virtual model continuously mirrors real-world soil conditions and provides actionable insights such as irrigation recommendations and soil health status.



Component	Quantity	Function
Soil Moisture Sensor	1	Measures soil water content
pH Sensor	1	Measures soil acidity
NodeMCU	1	IoT communication
Arduino UNO	1	Control unit
Relay 5V	1	Pump switching
AC Water Pump	2	Irrigation
DHT11	1	Temp & Humidity

Figure.1. Project Work Flow

Table – 1. Hardware Components

Figure.2. Soil Moisture Sensor

The soil moisture sensor is responsible for measuring the volumetric water content present in the soil. It plays a critical role in determining irrigation requirements by identifying whether the soil is dry, optimal, or overwatered.

Figure.3. pH Sensor

The pH sensor measures the acidity or alkalinity of the soil, which directly affects nutrient availability and crop productivity. Maintaining an optimal pH range ensures better fertilizer absorption and healthier plant growth.

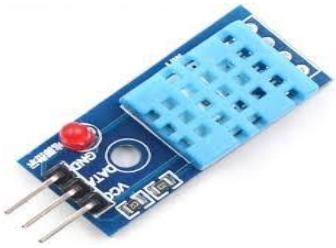


Figure.4. DHT11 Sensor

The DHT11 sensor captures atmospheric temperature and humidity surrounding the farmland. These environmental parameters influence soil evaporation rates and irrigation frequency. By integrating climate data into the Digital Twin model, the system improves prediction accuracy and supports smarter farming decisions.

Figure.4. NodeMCU ESP8266

NodeMCU acts as the primary IoT gateway of the system. It collects sensor readings and transmits them wirelessly to the web server using Wi-Fi connectivity. This enables remote monitoring and eliminates the need for manual field inspection.

Figure.5. Arduino Uno

The Arduino UNO serves as the control unit responsible for coordinating sensor inputs and actuator responses. It processes real-time data and activates the relay module whenever irrigation is required. Its reliable architecture ensures stable system performance even in continuous agricultural operation.

Figure.6. AC Water Pump

The AC water pumps are used to deliver water directly to the field when triggered by the control system. Automated pumping ensures timely irrigation, prevents crop stress, and reduces water wastage.

Software and Digital Twin Platform

In addition to the hardware infrastructure, the proposed system incorporates a web-based platform that acts as the Digital Twin of the agricultural field.

The website performs the following functions:

• Displays real-time soil data
• Generates historical graphs for trend analysis
• Provides irrigation suggestions
• Predicts soil conditions using machine learning
• Enables remote farm monitoring

The Digital Twin continuously synchronizes with sensor data, allowing farmers to visualize field conditions from anywhere. Historical analytics assist in identifying long-term soil behaviour patterns, improving crop planning and resource allocation.

IV. METHODOLOGY

The methodology describes the systematic procedure adopted to design and implement the proposed Soil Digital Twin System for Smart Farming Applications using IoT and Machine Learning. The framework integrates sensor-based data acquisition, intelligent data processing, virtual soil modelling, predictive analytics, and automated irrigation control to enable precision agriculture.

The overall methodological workflow ensures continuous synchronization between the physical soil environment and its virtual counterpart, allowing farmers to make data-driven decisions while minimizing manual intervention.

A. Soil Data Acquisition

Accurate soil monitoring forms the foundation of the proposed system. Multiple sensors are deployed at strategically selected locations within the agricultural field to capture essential soil and environmental parameters. The soil moisture sensor measures water content to determine irrigation requirements, while the pH sensor evaluates soil chemical properties that influence nutrient availability. Additionally, the DHT11 sensor records temperature and humidity, enabling the system to account for environmental factors affecting soil behaviour.

Sensor readings are continuously collected and transmitted to the microcontroller, ensuring real-time visibility into field conditions. This approach reduces dependency on traditional manual inspection methods and enhances monitoring accuracy.

B. Data Transmission and IoT Integration

After acquisition, the sensed data is processed by the Arduino UNO, which acts as the primary control unit. The processed data is then transmitted to the cloud server through the NodeMCU module using wireless communication. The IoT-enabled architecture facilitates remote accessibility, allowing farmers to monitor soil conditions from any location through the web interface. Continuous data streaming also ensures that the Digital Twin remains updated with real-world soil dynamics. This communication framework improves operational efficiency and eliminates delays associated with manual data collection.

C. Data Preprocessing and Storage

Raw sensor data often contains noise, fluctuations, or incomplete readings caused by environmental disturbances or hardware limitations. Therefore, pre-processing techniques are applied before analytical modelling.

The pre-processing phase includes:

• Removal of inconsistent readings
• Handling missing values
• Data normalization
• Outlier detection

The refined dataset is stored in a centralized database where it becomes available for historical analysis and machine learning operations. Maintaining structured data improves model reliability and supports long-term agricultural insights.

D. Digital Twin Model Development

The core innovation of the proposed system lies in the creation of a Digital Twin a dynamic virtual replica of the physical soil environment. Using real-time sensor inputs, the Digital Twin continuously mirrors soil conditions on the web platform. The virtual model provides graphical representations, parameter dashboards, and historical trend visualization to enhance situational awareness.

By simulating soil behaviour under varying climatic and irrigation conditions, the Digital Twin enables proactive farm management rather than reactive decision-making. This virtual representation significantly improves resource planning and reduces the risk of crop stress.

E. Machine Learning-Based Predictive Analytics

Machine learning techniques are incorporated to transform raw agricultural data into actionable intelligence. Historical and live sensor datasets are analysed to identify patterns in soil moisture variation, temperature shifts, and environmental changes.

Predictive models such as regression and time-series forecasting are employed to estimate future soil conditions and recommend optimal irrigation schedules.

The predictive capability allows farmers to:

• Prevent over-irrigation
• Reduce water wastage
• Maintain soil fertility
• Improve crop yield

As new data is continuously received, the models are updated to enhance prediction accuracy and system adaptability.

F. Automated Irrigation Control

To support precision farming, the system integrates an automated irrigation mechanism. When the moisture level falls below a predefined threshold, the Arduino activates the relay module, which in turn switches on the AC water pump. Once adequate moisture is restored, the pump is automatically deactivated, preventing excessive watering. This closed-loop control strategy ensures efficient water utilization while reducing farmer workload.

G. Web-Based Visualization and Decision Support

A dedicated web application serves as the visualization layer of the system. It functions as the interface of the Digital Twin and provides:

• Real-time soil parameter display
• Historical graphs
• Predictive insights
• Irrigation recommendations

The dashboard enables farmers to interpret complex sensor data easily and make informed agricultural decisions without requiring technical expertise.

H. System Evaluation

The proposed methodology is evaluated based on system responsiveness, prediction accuracy, reliability of sensor readings, and efficiency of automated irrigation.

Performance metrics include:

• Prediction Accuracy
• Mean Absolute Error (MAE)
• Root Mean Square Error (RMSE)
• System Latency
• Water Usage Efficiency

Testing under varying environmental conditions ensures that the system remains robust and suitable for real-world agricultural deployment.

V. RESULTS AND DISCUSSION

The proposed **Soil Digital Twin System for Smart Farming Applications using IoT & Machine Learning** was experimentally validated under real-time field conditions using integrated soil and environmental sensors. The system successfully acquired soil moisture, temperature, humidity, and pH data through the NodeMCU and Arduino-based sensing unit. The collected sensor data were transmitted wirelessly to the web-based digital twin platform, where real-time visualization and intelligent recommendations were generated.

Figure.7. Digital Twin Dashboard

The Digital Twin dashboard (Fig. 7) demonstrates real-time synchronization between the physical soil environment and its virtual replica. The grid-based visualization model effectively represents field conditions spatially, enabling farmers to observe moisture distribution across different sections. The system status indicator confirmed stable connectivity with a synchronization accuracy of approximately 100%.

The intelligence module analysed live sensor inputs and generated contextual recommendations. For example, when soil moisture was measured at 41.2%, the system correctly categorized the irrigation requirement as “Low” and suggested postponing watering to prevent over-irrigation. This demonstrates the capability of the decision-support engine to reduce unnecessary water usage while maintaining optimal soil health.

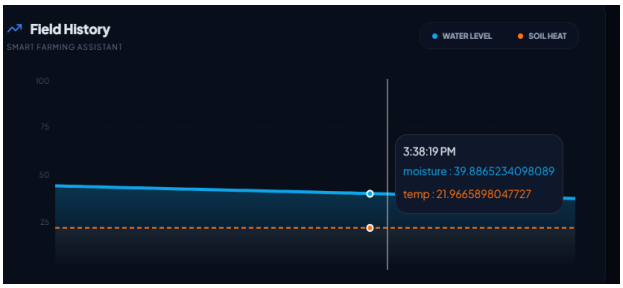


Figure.8. Historical Data

The historical data visualization module (Fig. 8) shows time-series plots of soil moisture and temperature. Experimental observations indicate that moisture levels decrease gradually during dry intervals and increase after irrigation cycles. Temperature trends remained stable within optimal crop growth range (22°C–30°C). The graphical interface enables trend analysis and supports predictive irrigation planning.

The relay-controlled water pump mechanism was tested under threshold-based automation conditions. When soil

moisture dropped below the predefined limit, the system successfully activated the AC water pumping motor. Upon reaching optimal moisture level, the relay deactivated automatically, confirming correct actuation performance.

Overall system evaluation confirms:

• Accurate real-time sensing
• Reliable wireless data transmission
• Stable Digital Twin synchronization
• Effective decision support
• Automated irrigation control

Compared to traditional manual monitoring methods, the proposed system significantly improves monitoring accuracy, reduces water wastage, and enhances farm-level decision-making efficiency.

Feature	Manual Irrigation System	Proposed Soil Digital Twin System
Soil Monitoring	Manual observation	Real-time sensor-based monitoring
Data Recording	No historical storage	Cloud-based data storage
Decision Making	Based on farmer experience	Data-driven intelligent recommendations
Irrigation Control	Manual ON/OFF operation	Automatic relay-controlled irrigation
Water Utilization	High water wastage	Optimized water usage
Remote Monitoring	Not possible	Web-based remote access
Prediction Capability	Not available	Machine learning-based prediction
Response Time	Slow and delayed	Real-time response (< 2 sec)
Scalability	Limited	Easily scalable with IoT nodes
Accuracy	Low	High (sensor-based measurement)

Table-2 Comparison Between Manual Irrigation and Proposed Soil Digital Twin System

VI. CONCLUSION

This paper presented the design and implementation of a Soil Digital Twin System integrating IoT-based soil monitoring with a web-based intelligent decision-support platform. The system combines soil moisture, pH, temperature, and humidity sensors with cloud connectivity to create a real-time virtual replica of the agricultural field. Experimental results demonstrate reliable sensor performance, accurate data visualization, and effective automated irrigation control. The Digital Twin model enables farmers to monitor soil conditions remotely, analyse historical trends, and receive intelligent recommendations for irrigation and soil management.

The integration of IoT sensing, automation, and digital twin visualization enhances resource efficiency, reduces water wastage, and supports sustainable farming practices. The proposed system represents a scalable and practical solution for precision agriculture and smart farming applications.

VII. FUTURE SCOPE

Although the proposed system demonstrates reliable performance, several enhancements can further improve scalability and intelligence:

• Integration of machine learning models for predictive irrigation scheduling
• Crop-specific recommendation engine based on soil type and growth stage
• Multi-node distributed sensor deployment for large farms
• Integration of weather forecasting APIs for evapotranspiration-based irrigation control
• Cloud-based data analytics and mobile application integration
• Nutrient deficiency detection using additional NPK sensors
• Autonomous irrigation optimization using reinforcement learning

Future work will focus on developing a predictive-prescriptive digital twin capable of adaptive irrigation control across varying environmental conditions.

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